Hadoop Word Count Using MapReduce

1. Objective

To create a text file, store it in HDFS, execute the Word Count MapReduce program using Hadoop, and view the word frequency results.

2. Procedure

Step 1: Create a New Text File

touch prac.txt

Data Science is the process of collecting, analyzing, and interpreting data to discover useful insights and support decision-making. It combines statistics, programming, machine learning, and domain knowledge to solve real-world problems. Data Science is widely used in industries such as healthcare, finance, marketing, and technology to improve efficiency and predict future trends.

Step 2: Verify the File Contents

cat prac.txt

Step 3: Upload the File to HDFS

hdfsdfs -put prac.txt /demo

Step 4: Execute the Word Count Program

hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /demo /output

Step 5: Check the Output Directory

hdfsdfs -ls /output

Step 6: View the Word Count Results

hdfsdfs -cat /output/part-r-00000

3. Source Code

Input File (prac.txt)

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Commands Used

touch prac.txt

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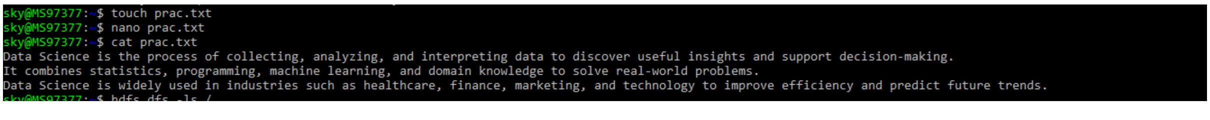
hdfsdfs -cat /output/part-r-00000

4. Screenshots

Creating the Input File



Verifying the File

```
sky@MS97377:~$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /demo /output
2026-07-26 23:31:05,223 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-26 23:31:06,263 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1785087544616_0001
2026-07-26 23:31:06,854 INFO input.FileInputFormat: Total input files to process : 1
2026-07-26 23:31:07,399 INFO mapreduce.JobSubmitter: number of splits:1
2026-07-26 23:31:07,696 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1785087544616_0001
2026-07-26 23:31:07,696 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-26 23:31:07,967 INFO conf.Configuration: resource-types.xml not found
2026-07-26 23:31:07,968 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-26 23:31:09,461 INFO impl.YarnClientImpl: Submitted application application_1785087544616_0001
2026-07-26 23:31:09,561 INFO mapreduce.Job: The url to track the job: http://MS97377.localdomain:8088/proxy/application_1785087544616_0001/
2026-07-26 23:31:09,563 INFO mapreduce.Job: Running job: job_1785087544616_0001
2026-07-26 23:31:38,649 INFO mapreduce.Job: Job job_1785087544616_0001 running in uber mode : false
2026-07-26 23:31:38,656 INFO mapreduce.Job:  map 0% reduce 0%
2026-07-26 23:31:48,992 INFO mapreduce.Job:  map 100% reduce 0%
2026-07-26 23:31:55,056 INFO mapreduce.Job:  map 100% reduce 100%
2026-07-26 23:31:56,109 INFO mapreduce.Job: Job job_1785087544616_0001 completed successfully
2026-07-26 23:31:56,383 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=611
                FILE: Number of bytes written=552689
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=487
                HDFS: Number of bytes written=433
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=7342
                Total time spent by all reduces in occupied slots (ms)=4136
                Total time spent by all map tasks (ms)=7342
                Total time spent by all reduce tasks (ms)=4136
                Total vcore-milliseconds taken by all map tasks=7342
                Total vcore-milliseconds taken by all reduce tasks=4136
                Total megabyte-milliseconds taken by all map tasks=7518208
                Total megabyte-milliseconds taken by all reduce tasks=4235264
```

Running Word Count

```
sky@MS97377:~$ hdfs dfs -ls /output
Found 2 items
-rw-r--r--   1 sky supergroup          0 2026-07-26 23:31 /output/_SUCCESS
-rw-r--r--   1 sky supergroup        433 2026-07-26 23:31 /output/part-r-00000
```

Screenshot 5: Checking Output Directory

```
sky@MS97377:~$ hdfs dfs -cat /output/part-r-00000
Data    2
It      1
Science 2
analyzing,      1
and     5
as      1
collecting,     1
combines        1
data    1
decision-making.        1
discover        1
domain  1
efficiency      1
finance,        1
future  1
healthcare,     1
improve 1
in      1
industries      1
insights        1
interpreting    1
is      2
knowledge       1
learning,       1
machine 1
marketing,      1
of      1
predict 1
problems.       1
process 1
programming,    1
real-world      1
solve   1
statistics,     1
such    1
support 1
technology      1
the     1
to      3
```

Screenshot 6: Viewing Word Count Results

```
sky@MS97377:~$ hdfs dfs -cat /wordcount_output/part-r-00000 | sort -t$'\t' -k2 -nr | head -3
the     3
of      3
is      3
sky@MS97377:~$
```

5. Conclusion

The Hadoop MapReduce Word Count application was successfully executed. The input text file was uploaded to HDFS, processed using the built-in Hadoop Word Count program, and the output containing the frequency of each word was generated and stored in the HDFS output directory. This experiment demonstrates the basic working of Hadoop MapReduce for distributed data processing.